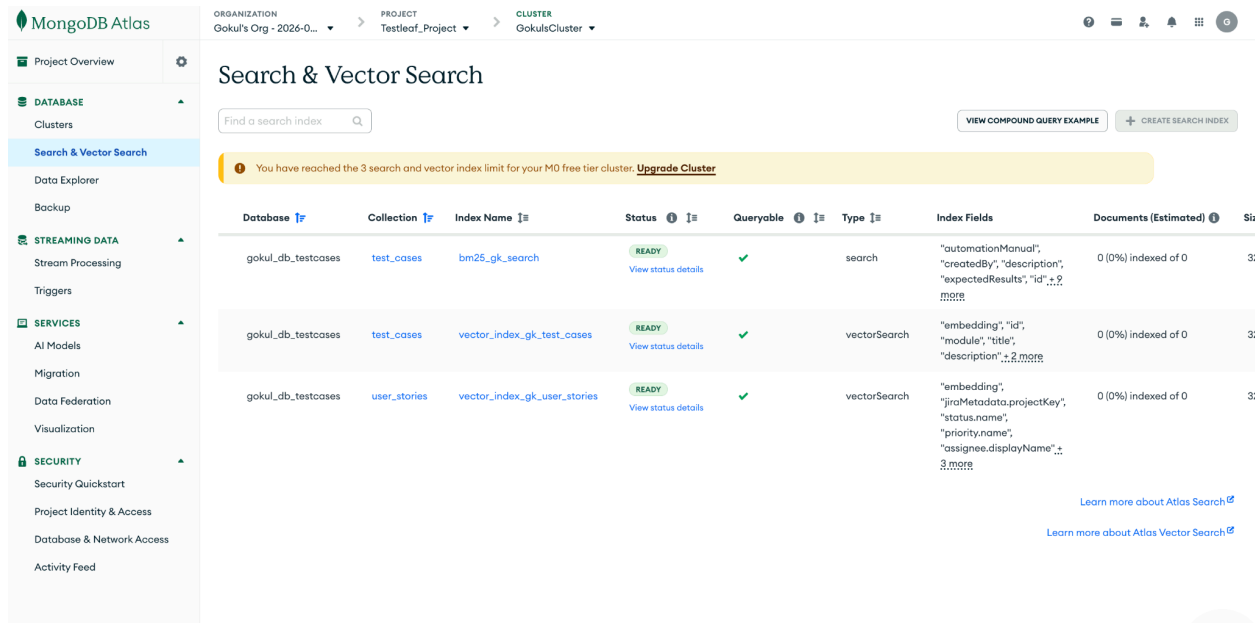


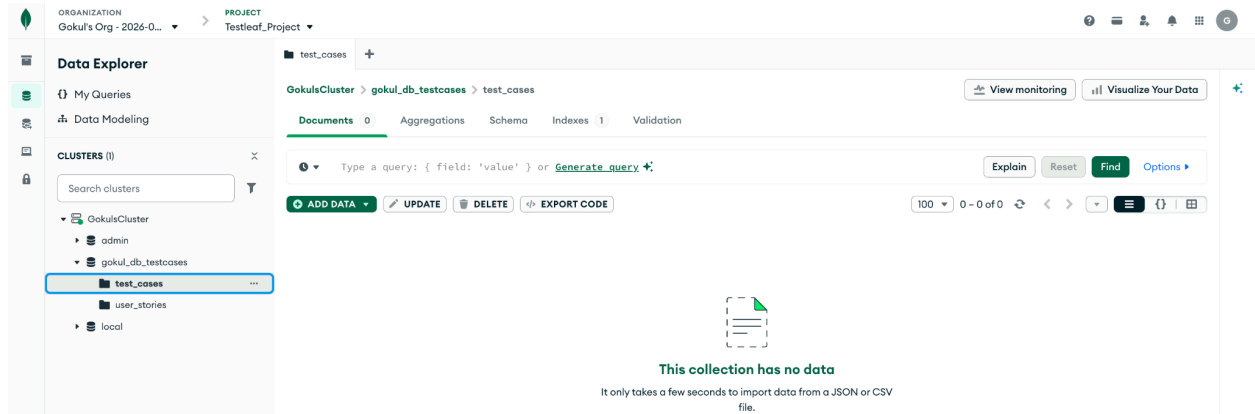
1.Existing Mongo DB Data Cleared: Vector & BM25 Search Cleared:



The screenshot shows the MongoDB Atlas interface for the 'Search & Vector Search' section. The left sidebar contains navigation options: Project Overview, DATABASE (Clusters), Search & Vector Search (selected), Data Explorer, Backup, STREAMING DATA (Stream Processing, Triggers), SERVICES (AI Models, Migration, Data Federation, Visualization), and SECURITY (Security Quickstart, Project Identity & Access, Database & Network Access, Activity Feed). The main content area displays a table of search indexes with columns: Database, Collection, Index Name, Status, Queryable, Type, Index Fields, Documents (Estimated), and Size. A yellow banner at the top indicates a limit on search and vector indexes for the M0 free tier cluster, with a link to 'Upgrade Cluster'. The table lists three indexes: 'bm25_gk_search' (search type), 'vector_index_gk_test_cases' (vectorSearch type), and 'vector_index_gk_user_stories' (vectorSearch type). All indexes are in a 'READY' status. Links to 'Learn more about Atlas Search' and 'Learn more about Atlas Vector Search' are provided at the bottom right.

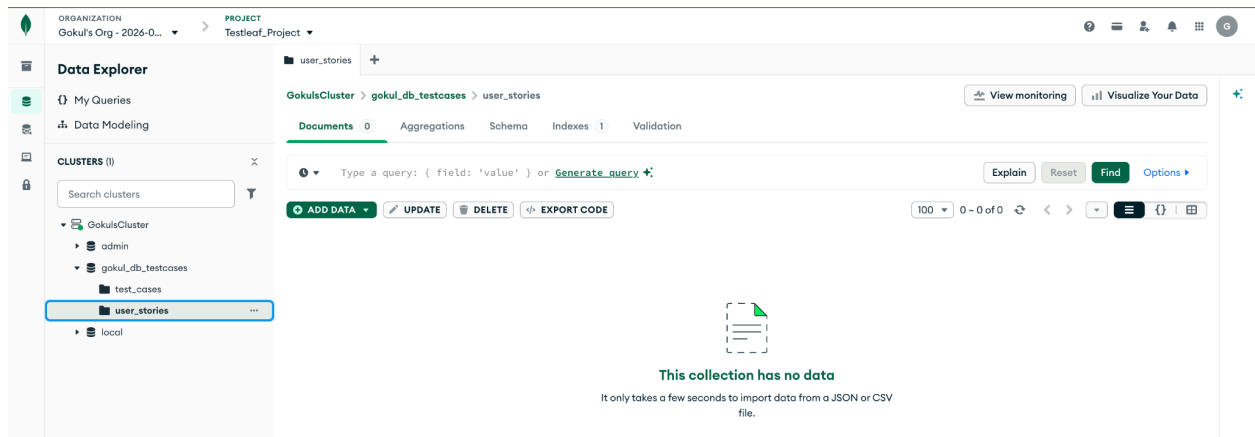
Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size
gokul_db_testcases	test_cases	bm25_gk_search	READY	✓	search	"automationManual", "createdBy", "description", "expectedResults", "id", "more	0 (0%) indexed of 0	3
gokul_db_testcases	test_cases	vector_index_gk_test_cases	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description", "more	0 (0%) indexed of 0	3
gokul_db_testcases	user_stories	vector_index_gk_user_stories	READY	✓	vectorSearch	"embedding", "jiraMetadata.projectKey", "status.name", "priority.name", "assignee.displayName", "more	0 (0%) indexed of 0	3

Test Cases in DB Cleared:



The screenshot shows the MongoDB Atlas Data Explorer interface. The left sidebar includes 'Data Explorer' (My Queries, Data Modeling), 'CLUSTERS' (GokulsCluster, admin, gokul_db_testcases, test_cases (selected), user_stories, local), and 'Security'. The main content area displays the 'test_cases' collection in the 'gokul_db_testcases' database. The 'Documents' tab is active, showing 0 documents. A message states: 'This collection has no data. It only takes a few seconds to import data from a JSON or CSV file.' The interface includes buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE', along with a search bar and navigation controls.

User Stories in DB Cleared:



The screenshot shows the MongoDB Atlas Data Explorer interface for the 'user_stories' collection. The left sidebar is identical to the previous screenshot, with 'user_stories' selected under 'CLUSTERS'. The main content area displays the 'user_stories' collection in the 'gokul_db_testcases' database. The 'Documents' tab is active, showing 0 documents. A message states: 'This collection has no data. It only takes a few seconds to import data from a JSON or CSV file.' The interface includes buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE', along with a search bar and navigation controls.

2.Ingested new test cases related to my car rental app domain: Converted test cases to JSON

The screenshot displays the 'RAG Pipeline' interface with the 'Convert to JSON' step selected. The left sidebar contains a navigation menu with options: 'Convert to JSON' (highlighted), 'Embeddings & Store', 'Query Preprocessing', 'Vector Search', 'BM25 Search', 'Hybrid Search', 'Score Fusion', 'Summarize & Dedup', and 'Prompt & Schema'. The main content area shows the 'Convert to JSON' step with a sub-header 'Upload and convert Excel files (Test Cases or User Stories)'. A table lists columns: 'Last modified date', 'Risk', 'Version', and 'Type'. Below the table, a 'Conversion Results' section shows a green checkmark and the message 'Success! Test cases file converted successfully'. It includes an 'Output File' section with the file 'converted-1779697384540.json' and a 'Status' section with a green 'Completed' button. A 'Conversion Log' section shows a warning for a missing sheet and a success message for converting 104 rows.

Navigation

- Convert to JSON
Upload and convert Excel files (Test Cases or User Stories)
- Embeddings & Store
Create and manage embeddings
- Query Preprocessing
Transform & expand queries
- Vector Search
Semantic vector search
- BM25 Search
Keyword-based search
- Hybrid Search
Combined BM25 + Vector
- Score Fusion
BM25+Vector fusion reranking
- Summarize & Dedup
AI summarization & deduplication
- Prompt & Schema
Configure prompt templates & JSON schemas

RAG Pipeline

Home > Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Last modified date	Risk	Version	Type
--------------------	------	---------	------

Conversion Results

Success! Test cases file converted successfully

Output File
converted-1779697384540.json
Saved to data folder

Status
Completed
Ready for embedding creation

Conversion Log

Sheet "testcases" not found. Using first sheet: "Sheet1"
Converted 104 rows from "Sheet1" into /Users/karthikradhakrishnan/Downloads/GokuL/Week8/D1/rag-mongo-demo-v9/src/data/converted-1779697384540.json

Embedding created successfully:

The screenshot displays the 'RAG Pipeline' interface with the 'Embeddings & Store' step selected. The left sidebar contains a navigation menu with options: 'Convert to JSON', 'Embeddings & Store' (highlighted), 'Query Preprocessing', 'Vector Search', 'BM25 Search', 'Hybrid Search', 'Score Fusion', 'Summarize & Dedup', and 'Prompt & Schema'. The main content area shows the 'Embeddings & Store' step with a sub-header 'Create and manage embeddings'. It includes a 'Button Disabled: false' and 'Files loaded: 9' section. A 'Processing Time' section shows '~10-30 seconds per file' and '~\$0.0001 per test case'. A 'Selected Files' section lists 'converted-1779697384540.json'. A 'Processing Results' section shows '1 completed, 0 failed' and a detailed log for 'testcases' completed, including input file, processing time, and configuration.

Navigation

- Convert to JSON
Upload and convert Excel files (Test Cases or User Stories)
- Embeddings & Store
Create and manage embeddings
- Query Preprocessing
Transform & expand queries
- Vector Search
Semantic vector search
- BM25 Search
Keyword-based search
- Hybrid Search
Combined BM25 + Vector
- Score Fusion
BM25+Vector fusion reranking
- Summarize & Dedup
AI summarization & deduplication
- Prompt & Schema
Configure prompt templates & JSON schemas

RAG Pipeline

Home > Embeddings & Store

Create and manage embeddings

Button Disabled: false
Files loaded: 9

TEST SELECTION SELECT FIRST FILE

Processing Time: ~10-30 seconds per file
Cost: ~\$0.0001 per test case

Selected Files

converted-1779697384540.json

Processing Results
1 completed, 0 failed

testcases completed

Input file: src/data/converted-1779697384540.json
MISTRAL AI BATCH PROCESSING: 104 test cases
Using Mistral AI Embedding API for Cost-Effective Processing!
Configuration:
Batch Size:...

Vector search & BM25 created successfully in DB for new 104 test cases:

The screenshot shows the 'Search & Vector Search' interface. On the left is a sidebar with navigation options: Search & Vector Search, AUTO EMBEDDING, Usage, Rate Limits, SHORTCUTS, Cluster Metrics, and Data Explorer. The main area has a header with 'Search & Vector Search' and a search bar. Below the header is a table listing search indexes.

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size
gokul_db_testcases	test_cases	bm25_gk_search	READY	✓	search	"automationManual", "createdBy", "description", "expectedResults", "id", "more"	104 (100%) indexed of 104	324B
gokul_db_testcases	test_cases	vector_index_gk_test_cases	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description", "more"	104 (100%) indexed of 104	325B

User stories to JSON:

The screenshot shows the 'RAG Pipeline' interface, specifically the 'Convert to JSON' step. The left sidebar contains navigation options: Convert to JSON, Embeddings & Store, Query Preprocessing, Vector Search, BM25 Search, Hybrid Search, Score Fusion, Summarize & Dedup, and Prompt & Schema. The main area has a header with 'RAG Pipeline' and a sub-header 'Convert to JSON'. Below the header is a form for uploading and converting Excel files.

File Upload
Select data type and configure your Excel file

Data Type: **US** User Stories

SELECT EXCEL FILE (USER STORIES)

Gokul_Domain_User_Stories.xlsx 10.4 KB **Selected**

Sheet Name: **stories**

Name of the Excel sheet to convert (default: stories)

CONVERT USER STORIES **RESET**

Expected User Stories Columns

Your Excel file should contain these columns:

- Story ID / Key
- Summary / Title
- Description
- Status
- Priority
- Assignee
- Reporter
- Story Points
- Components
- Labels
- Sprint
- Epic
- Acceptance Criteria
- Created Date
- Updated Date

File selected successfully

The screenshot shows the 'Conversion Results' interface. The left sidebar contains navigation options: Query Preprocessing, Vector Search, BM25 Search, Hybrid Search, Score Fusion, Summarize & Dedup, and Prompt & Schema. The main area has a header with 'Conversion Results' and a sub-header 'Success! User stories file converted successfully'. Below the header is a table showing the output file and status.

Conversion Results

Success! User stories file converted successfully

Output File	Status
stories-1779697677178.json Saved to data folder	Completed Ready for embedding creation

Conversion Log

- ⚠ Sheet "stories" not found. Using first sheet: "Sheet1"
- 🔄 Converting 38 rows from sheet "Sheet1"...
- ✅ Converted 38 user stories from "Sheet1" into /Users/karthikradhakrishnan/Downloads/Gokul/Week8/D1/rag-mongo-demo-v9/src/data/stories-1779697677178.json
- ⚠ Filtered out: 0 empty rows

Embedding created for new user stories:

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

RAG Pipeline

Home > Embeddings & Store

Create and manage embeddings

☐	stories-1770988242140.json	198.29 KB	13/02/2026 18:40:44	JSON
☐	stories-1770988363821.json	198.29 KB	13/02/2026 18:42:45	JSON
☐	stories-1775982680953.json	198.29 KB	12/04/2026 14:01:21	JSON
☒	stories-1779697677178.json	40.29 KB	25/05/2026 13:57:57	JSON

1 row selected

Rows per page: 10 1-10 of 10

Selected 1 file(s): stories-1779697677178.json

Processing Embeddings...

0/1

Current file: user-stories

Progress: 0%

Job ID: job-1779697828575-qv...

This process continues even if you refresh the page!

Batch processing started! Job ID: job-1779697828575-qvjoss1wk

Starting batch processing for 1 files...

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

RAG Pipeline

Home > Embeddings & Store

Create and manage embeddings

Button Disabled: false

Files loaded: 10

TEST SELECTION

SELECT FIRST FILE

Processing Time: ~10-30 seconds per file

Cost: ~\$0.0001 per test case

Selected Files

stories-1779697677178.json

Processing Results

1 completed, 0 failed

user-stories

completed

Input file: src/data/stories-1779697677178.json

MISTRAL AI BATCH PROCESSING: 38 user stories

Using Mistral AI Embedding API for Cost-Effective Processing!

Configuration:

Batch Size: ...

Vector search updated for user stories:

The screenshot shows the MongoDB Atlas Search & Vector Search interface. The left sidebar contains navigation options: Project Overview, DATABASE (Clusters), Search & Vector Search (selected), Data Explorer, Backup, STREAMING DATA (Stream Processing, Triggers), SERVICES (AI Models, Migration, Data Federation, Visualization), SECURITY (Security Quickstart, Project Identity & Access, Database & Network Access, Activity Feed), and a top navigation bar with Organization, Project, and Cluster dropdowns.

The main content area is titled "Search & Vector Search" and includes a search bar and buttons for "VIEW COMPOUND QUERY EXAMPLE" and "+ CREATE SEARCH INDEX". A yellow banner states: "You have reached the 3 search and vector index limit for your M0 free tier cluster. [Upgrade Cluster](#)".

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size
gokul_db_testcases	test_cases	bm25_gk_search	READY View status details	✓	search	"automationManual", "createdBy", "description", "expectedResults", "id", "module"	104 (100%) indexed of 104	73.6KB
gokul_db_testcases	test_cases	vector_index_gk_test_cases	READY View status details	✓	vectorSearch	"embedding", "id", "module", "title", "description", "expectedResults"	104 (100%) indexed of 104	500.5KB
gokul_db_testcases	user_stories	vector_index_gk_user_stories	READY View status details	✓	vectorSearch	"embedding", "jiraMetadata.projectKey", "status.name", "priority.name", "assignee.displayName", "description"	38 (100%) indexed of 38	326B

Links at the bottom right: [Learn more about Atlas Search](#) and [Learn more about Atlas Vector Search](#).

New test cases and user stories in DB

The top screenshot shows the Data Explorer for the "test_cases" collection. The left sidebar lists "test_cases" under the "gokul_db_testcases" database. The main view displays the "Documents" tab with 104 documents. A query is entered: `{ field: 'value' }` or [Generate query](#). The "Find" button is highlighted. Below the query, a sample document is shown:

```
{
  "_id": ObjectId("6a146751fed7da893add8e6"),
  "module": "Login",
  "id": "TC_001",
  "preRequisites": "Valid user account exists",
  "title": "Verify valid user login",
  "description": "Validate successful login into the application",
  "steps": "1. Launch app\n\n2. Enter valid username\n\n3. Enter password\n\n4. Tap L-\"",
  "expectedResults": "User should successfully login to the app",
  "automationManual": "Automated"
}
```

The bottom screenshot shows the Data Explorer for the "user_stories" collection. The left sidebar lists "user_stories" under the "gokul_db_testcases" database. The main view displays the "Documents" tab with 38 documents. A query is entered: `{ field: 'value' }` or [Generate query](#). The "Find" button is highlighted. Below the query, a sample document is shown:

```
{
  "_id": ObjectId("6a1408aaafcc5bf816d74023"),
  "key": "BAT-001",
  "summary": "User Login Functionality",
  "description": "As a registered user, I want to login using valid credentials so that ...",
  "status": Object,
  "priority": Object,
  "assignee": null,
  "reporter": null,
  "created": "2026-01-05T00:00:00.000Z",
  "updated": "2026-05-25T08:27:57.288Z",
  "components": Array (empty)
}
```

3. Vector Search, BM25 Search & Hybrid Search Testing: Vector:

Navigation

Convert to JSON

Upload and convert Excel files
(Test Cases or User Stories)

Embeddings & Store

Create and manage
embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization &
deduplication

Prompt & Schema

Configure prompt templates &
JSON schemas

RAG Pipeline

Semantic vector search

Home > Vector Search

Query Search

Search through your test cases using semantic similarity with AI embeddings.

Search Query

Enter your search terms and find similar test cases

Enter your search query

app login and vehicle search

Use descriptive terms related to test functionality

Advanced Metadata Filters

SHOW FILTERS

Results Limit

5 Results

SEARCH

Search Query: "app login and vehicle search"

Performance: \$0.000001 cost 7 tokens

5 results

Search Results

5 results ranked by semantic similarity

Test Case	Similarity	Description	Source
Verify search by city	86.3%	Validate vehicle search by city/location	Vehicle Search
Verify nearby search functionality	85.9%	Validate search using current location	Vehicle Search
Verify search filter by vehicle type	85.5%	Validate filtering based on SUV/Sedan	Vehicle Search
Verify search analytics tracking	84.9%	Validate analytics event generation	Vehicle Search

Rows per page: 10 1-5 of 5

Detailed View

Verify search by city

Score: 0.8633

Vehicle Search

Verify nearby search functionality

Score: 0.8595

Vehicle Search

Verify search filter by vehicle type

Score: 0.8552

Vehicle Search

Verify search analytics tracking

Score: 0.8491

Vehicle Search

Verify recommendation relevance

Score: 0.8480

Vehicle Selection

BM25:

Navigation

Convert to JSON

Upload and convert Excel files
(Test Cases or User Stories)

Embeddings & Store

Create and manage
embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization &
deduplication

Prompt & Schema

Configure prompt templates &
JSON schemas

RAG Pipeline

Home > BM25 Search

Keyword-based search

BM25 Keyword Search

Fast keyword-based search using BM25 algorithm. Best for exact matches, IDs, and specific terms.

Search Query

app login and vehicle search

Result...

10

SEARCH

SHOW FILTERS

BM25 Search Results: Found 10 test cases in 93ms

Search Type: bm25

Search Results (10)

#1 TC_001

Score: 8.34

Verify valid user login

Module: Login

P1

Risk: High

Automated

Description: Validate successful login into the application

Steps: 1. Launch app 2. Enter valid username 3. Enter password 4. Tap login

Navigation

Convert to JSON

Upload and convert Excel files
(Test Cases or User Stories)

Embeddings & Store

Create and manage
embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization &
deduplication

Prompt & Schema

Configure prompt templates &
JSON schemas

RAG Pipeline

Home > BM25 Search

Keyword-based search

Created: 25/05/2026

#9 TC_073

Score: 5.48

Verify multi-device login

Module: Login

P2

Risk: Medium

Manual

Description: Validate simultaneous login behavior

Steps: 1. Login on two devices

Expected Results: System should handle sessions correctly

Created: 25/05/2026

#10 TC_063

Score: 5.40

Verify login without internet

Module: Login

P1

Risk: High

Automated

Description: Validate offline login behavior

Steps: 1. Disable internet 2. Attempt login

Expected Results: Appropriate offline message should display

Created: 25/05/2026

Hybrid:Balanced:

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Hybrid Search Results: Found 10 test cases in 3982ms

Stats: 10 in both indexes,0 BM25 only,0 vector only

Timing: BM25: 56ms, Vector: 3923ms

Cost: \$0.000001 (7 tokens)

Search Results (10)

#1 TC_001

Hybrid: 91.7%

Found in: both

Verify valid user login

Module: Login P1 Risk: High Automated

Description: Validate successful login into the application

Steps: 1. Launch app 2. Enter valid username 3. Enter password 4. Tap Login

Expected Results: User should successfully login to the app

Score Breakdown:

BM25: 100.0% (raw: 8.34) Vector: 83.4% (raw: 0.83) Hybrid: 91.7%

Created: 25/05/2026

#2 TC_064

Hybrid: 86.4%

Found in: both

Verify search under poor network

Keyword:

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Hybrid Search Results: Found 10 test cases in 4198ms

Stats: 10 in both indexes,0 BM25 only,0 vector only

Timing: BM25: 52ms, Vector: 4144ms

Cost: \$0.000001 (7 tokens)

Search Results (10)

#1 TC_001

Hybrid: 95.0%

Found in: both

Verify valid user login

Module: Login P1 Risk: High Automated

Description: Validate successful login into the application

Steps: 1. Launch app 2. Enter valid username 3. Enter password 4. Tap Login

Expected Results: User should successfully login to the app

Score Breakdown:

BM25: 100.0% (raw: 8.34) Vector: 83.4% (raw: 0.83) Hybrid: 95.0%

Created: 25/05/2026

#2 TC_064

Hybrid: 87.3%

Found in: both

Verify search under poor network

Semantic:

Upload and convert Excel files

(Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Hybrid Search Results: Found 10 test cases in 4071ms

Stats: 10 in both indexes,0 BM25 only,0 vector only

Timing: BM25: 51ms, Vector: 4760ms

Cost: \$0.000001 (7 tokens)

Search Results (10)

#1 TC_001

Hybrid: 88.4%

Found in: both

Verify valid user login

Module: Login P1 Risk: High Automated

Description: Validate successful login into the application

Steps: 1. Launch app 2. Enter valid username 3. Enter password 4. Tap Login

Expected Results: User should successfully login to the app

Score Breakdown:

BM25: 100.0% (raw: 8.34) Vector: 83.4% (raw: 0.83) Hybrid: 88.4%

Created: 25/05/2026

#2 TC_064

Hybrid: 85.6%

Found in: both

Verify search under poor network

4. Reranking, Deduplication & Summarization:

Reranking After:

Vector Search
Semantic vector search

BM25 Search
Keyword-based search

Hybrid Search
Combined BM25 + Vector

Score Fusion
BM25+Vector fusion reranking

Summarize & Dedup
AI summarization & deduplication

Prompt & Schema
Configure prompt templates & JSON schemas

Settings

AFTER RERANKING (FINAL)BEFORE RERANKING (ORIGINAL)COMPARISON TABLE

After Groq AI Reranking (10 results)

#1TC_001Rerank Score: 0.0000

Verify valid user login

Module: LoginP1Risk: High

Description: Validate successful login into the application

Ranking Scores:
BM25 Score: 8.3430Groq Rerank: 0.5000

#2TC_033Rerank Score: 0.0000

Reranking Before:

Vector Search
Semantic vector search

BM25 Search
Keyword-based search

Hybrid Search
Combined BM25 + Vector

Score Fusion
BM25+Vector fusion reranking

Summarize & Dedup
AI summarization & deduplication

Prompt & Schema
Configure prompt templates & JSON schemas

AFTER RERANKING (FINAL)BEFORE RERANKING (ORIGINAL)COMPARISON TABLE

Initial Candidate Ranking (Before Groq AI Reranking - 10 results from 50 candidates)

These are the top candidates from the 50-candidate pool before Groq AI semantic reranking. Notice how Groq reorders them based on actual relevance to your query!

Candidate Rank: #1TC_001Vector: 0.0000

Verify valid user login

Module: LoginP1

Description: Validate successful login into the application

Candidate Rank: #2TC_033Vector: 0.0000

Comparison table:

Before vs After Comparison

Original Rank (Candidates)	Test Case ID	Title	Vector Score	Groq Rerank Score	Final Rank
#?	TC_001	Verify valid user login	0.0000	0.5000	#1
#?	TC_033	Verify biometric login	0.0000	0.5000	#2
#?	TC_064	Verify search under poor network	0.0000	0.5000	#3
#?	TC_004	Verify search by city	0.0000	0.5000	#4
#?	TC_084	Verify search analytics tracking	0.0000	0.5000	#5
#?	TC_005	Verify nearby search functionality	0.0000	0.5000	#6
#?	TC_036	Verify search filter by vehicle type	0.0000	0.5000	#7
#?	TC_074	Verify multilingual search UI	0.0000	0.5000	#8
#?	TC_073	Verify multi-device login	0.0000	0.5000	#9
#?	TC_063	Verify login without internet	0.0000	0.5000	#10

Deduplication:

(Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Demo v1.2

Enterprise Edition

<

RAG Pipeline

Home

> Summarize & Dedup

AI summarization & deduplication

Search Summarization & Deduplication

Search, remove duplicates, and generate AI-powered summaries of test cases

Search Query

app login and vehicle search

Search Type

Vector

Limit

20

Quick Examples:

patient login authentication

merge UHID records

appointment booking

password reset OTP

SEARCH

DEDUPLICATE

SUMMARIZE WITH AI

COPY ALL RESULTS

Dedup Threshold

0.85

Summary Type

Concise

SEARCH RESULTS

20

DEDEDUPICATED

AI SUMMARY

Original Results (20)

COPY ALL

TC_004

Score: 0.8633

Verify search by city

Validate vehicle search by city/location

Vehicle Search

P1 Priority

Functional

TC_005

Score: 0.8595

Verify nearby search functionality

Validate search using current location

Vehicle Search

P2 Priority

Integration

(Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Demo v1.2

Enterprise Edition

<

RAG Pipeline

Home

> Summarize & Dedup

AI summarization & deduplication

Search Summarization & Deduplication

Search, remove duplicates, and generate AI-powered summaries of test cases

Search Query

app login and vehicle search

Search Type

Vector

Limit

20

Quick Examples:

patient login authentication

merge UHID records

appointment booking

password reset OTP

SEARCH

DEDUPLICATE

SUMMARIZE WITH AI

COPY ALL RESULTS

Dedup Threshold

0.85

Summary Type

Concise

Show Duplicates

SEARCH RESULTS

20

DEDEDUPICATED

AI SUMMARY

Deduplication Statistics

Original Count

20

After Deduplication

20

Duplicates Removed

0

Reduction

0.0%

Unique Results (20)

COPY ALL UNIQUE

TC_004

Unique

Score: 0.8633

Verify search by city

TC_005

Unique

Score: 0.8595

Verify nearby search functionality

Summarization:

(Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

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Prompt & Schema

Configure prompt templates & JSON schemas

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Configure environment

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AI SUMMARY

AI-Generated Summary

Overview of Test-Case Search Results (300 words max)

The list contains 20 test cases focused primarily on the "Vehicle Search" module (11 cases, ≈ 55 % of the set), with additional coverage of "Vehicle Selection", "Vehicle Availability", and "Vehicle Booking". Relevance scores range from "86.3%" (highest) down to "83.6%", indicating a tightly clustered, high-priority set.

I Module | # of Cases | Core Focus |

|-----|-----|

I "Vehicle Search" | 11 | City-based search, nearby/voice search, filters (type, price, multilingual UI), analytics, recent searches, network resilience, duplicate handling, and search-under-poor-network behavior. |

I "Vehicle Selection" | 5 | Recommendation relevance & AI-driven suggestions, display of recommended vehicles, accessibility (screen-reader labels), image-gallery loading, and vehicle-details page validation. |

I "Vehicle Availability" | 4 | Availability display, handling of unavailable vehicles, multi-city inventory checks, and empty-inventory responses. |

I "Vehicle Booking" | 1 | AI-driven booking recommendation accuracy. |

(Test Cases or User Stories)

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language and network conditions.

2. "Filtering & Sorting" – Ensures filters (vehicle type) and sorting (price) work correctly.

3. "Analytics & Tracking" – Confirms that search actions generate proper analytics events.

4. "Recommendation Engine" – Checks relevance, uniqueness, and AI-driven suggestions across selection and booking flows.

5. "Availability & Edge Cases" – Tests display of available vs. unavailable vehicles, multi-city inventory, and graceful handling when inventory is empty.

6. "Accessibility & UI Quality" – Verifies screen-reader labels, image-gallery loading, and overall UI consistency.

Relevance Highlights

- The top-ranked case ("TC_004") verifies city-based search (86.3%).

- Nearby-search, vehicle-type filter, and analytics tracking follow closely (85.9%–84.9%).

- All remaining cases stay above 84 %, indicating a uniformly high priority across functional areas.

Takeaway: The collection provides a comprehensive, high-relevance test suite that covers core search capabilities, recommendation accuracy, availability handling, and user-experience aspects (accessibility, multilingual UI, performance under poor network). Prioritizing execution of the Vehicle Search cases will address the majority of critical search

Token Usage & Cost

Model	Prompt Tokens	Completion Tokens	Total Cost
openai/gpt-oss-120b	1,958	646	\$0.000000
	\$0.000000	\$0.000000	2,604 tokens

Pricing: Free with Groq API (llama-3.3-70b-versatile model)

5. Analyse and understanding:

a. According to my analysis reranking re-evaluates retrieved test cases and sorts them again based on deeper semantic relevance

User inputs query -> This goes to embedding search -> retrieves test cases and assigns rank for each test case -> Goes via reranker model -> Final output via of reranked test case for search relevance

b. Duplicate results are identified and removed using similarity comparison techniques.

Retrieve multiple test cases from DB -> Compare test cases via multiple methods -> if test cases are similar remove the duplicates

c. Vector Search retrieves results based on semantic meaning using embeddings. Query to vector embeddings -> compare query vector with document vector -> using cosine similarity retrieves similar results

BM25 retrieves results using keyword matching and term frequency

Match exact query with keywords -> Score test cases based on keyword frequency, document length and keyword importance

Hybrid Search combines Vector Search and BM25 Search

d. Hybrid Search, weightages determine how much importance is given to: Vector Search (semantic meaning) and BM25 Search (exact keyword match). The final ranking score is calculated by combining both scores.